Shivli Agrawal

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Education

Purdue University Jan 2022 – Dec 2025
Bachelor of Science, Computer Science *West Lafayette, IN*
- Dean's List: 2022-2023

Professional Experience

Undergraduate Teaching Assistant (For 7 terms) Aug 2022 – Present
Purdue University, Object Oriented Programming with JAVA guided by Prof. Dunsmore *West Lafayette, IN*
- Supervised lab sessions for **80+ students**, improving their programming skills and understanding of OOP.
- Debugged assignments, provided detailed feedback, and maintained accurate performance records over the past **2+ years**.

Software Development Intern June 2024 – Aug 2024
No Limit Living LLC, Guide: Emmani Ellis *Gary, IN*
- Enhanced the company's website using **Node.js** and **React.js**, achieving a **30% reduction in page load time.**
- Developed a mobile-friendly version and Android app using **React Native** and **Express.js**.
- Integrated APIs using Python libraries like Beautiful Soup, enabling content updates and improving website functionality. Performed **Unit** and **QA testing**, and maintained detailed documentation efforts.

Front-end Development Intern May 2023 – Aug 2023
Qnucleon PTY LTD *Remote*
- Transformed website aesthetics using HTML and JavaScript, modernizing its appearance to align with contemporary design increasing **click-through rates by 17%**.
- Conducted usability testing and implemented iterative improvements for a seamless user experience.

Data Science Researcher Aug 2022 – May 2023
Battelle Inc, Guide: Ujjwal Garg *West Lafayette, IN*
- Improved knowledge graph generation for drug-related health records using advanced **ML techniques**.
- Utilized **BERT** to extract information from a Harvard Medical School dataset on Adverse Drug Events (Track 2) and employed advanced data preprocessing techniques for HuggingFace friendly Dataset objects.
- Fine-tuned **BioBERT** with Hyperband and Population-based training, achieving a **5% increase** in named entity recognition (NER) performance.

Web Development Researcher Aug 2023 – Dec 2023
No Limit Living, LLC *West Lafayette, IN*
- Designed a comprehensive website supporting the company's daily operations and integrating internal MLS tools through The Data Mine Corporate Partnership at Purdue.
- Implemented live updates and analytics dashboards, improving operational efficiency.

Projects

Nocturnè Marketplace — MERN Stack (MongoDB, Express, React, Node.js), JWT, Cloudflare Feb 2025
- Built a **dynamic e-commerce platform** with category-based navigation, JWT authentication, and real-time search filtering.
- Optimized **API performance** with Cloudflare Workers and enhanced UI/UX with **CSS Grid, Flexbox**, and animation effects

System Monitor — C, GTK, Linux kernel, /proc file-system Nov 2024
- Developed a **GTK-based graphical tool** in C language to monitor CPU, memory usage, and active processes.
- Leveraged the **/proc file-system** for real-time data visualization and implemented process management features.

BoilerTutors — React.js, Node.js, PostgreSQL, REST APIs Sept 2024
- Developed a full-stack web app with **React.js** and **Node.js** for **resource sharing and collaboration.**
- Designed robust **REST APIs** for data management and used **PostgreSQL** for user and schedule management.

Database Learning Management System Apr 2022
- Designed a **GUI-based LMS** with concurrency support for quizzes, auto-grading, and real-time student-teacher interactions.

Technical Skills

- **Languages**: Java, C, C++, Python, R, JavaScript
- **Frameworks & Tools**: React.js, Node.js, Express.js, PyTorch, MongoDB, MySQL, PostgreSQL, Figma, Pandas
- **Technologies**: Machine Learning, Deep Learning, Natural Language Processing, High-Performance Computing (HPC), DevOps, Dynamic Programming.
- **Others**: Data Structures and Algorithms, Object-Oriented Programming, Git, Agile Methodology, Software Testing, Linux, Shell Scripting, Computer Architecture, Systems Programming, Software Engineering and Design

Leadership positions

President Aug 2023 – Present
UR Global, Purdue *West Lafayette, IN*
- Leading and organizing cultural events, promoting cross-cultural understanding, and fostering a welcoming environment for diverse members, enhancing global awareness and cultural exchange within the club and the broader community.